NeuroCogniAI

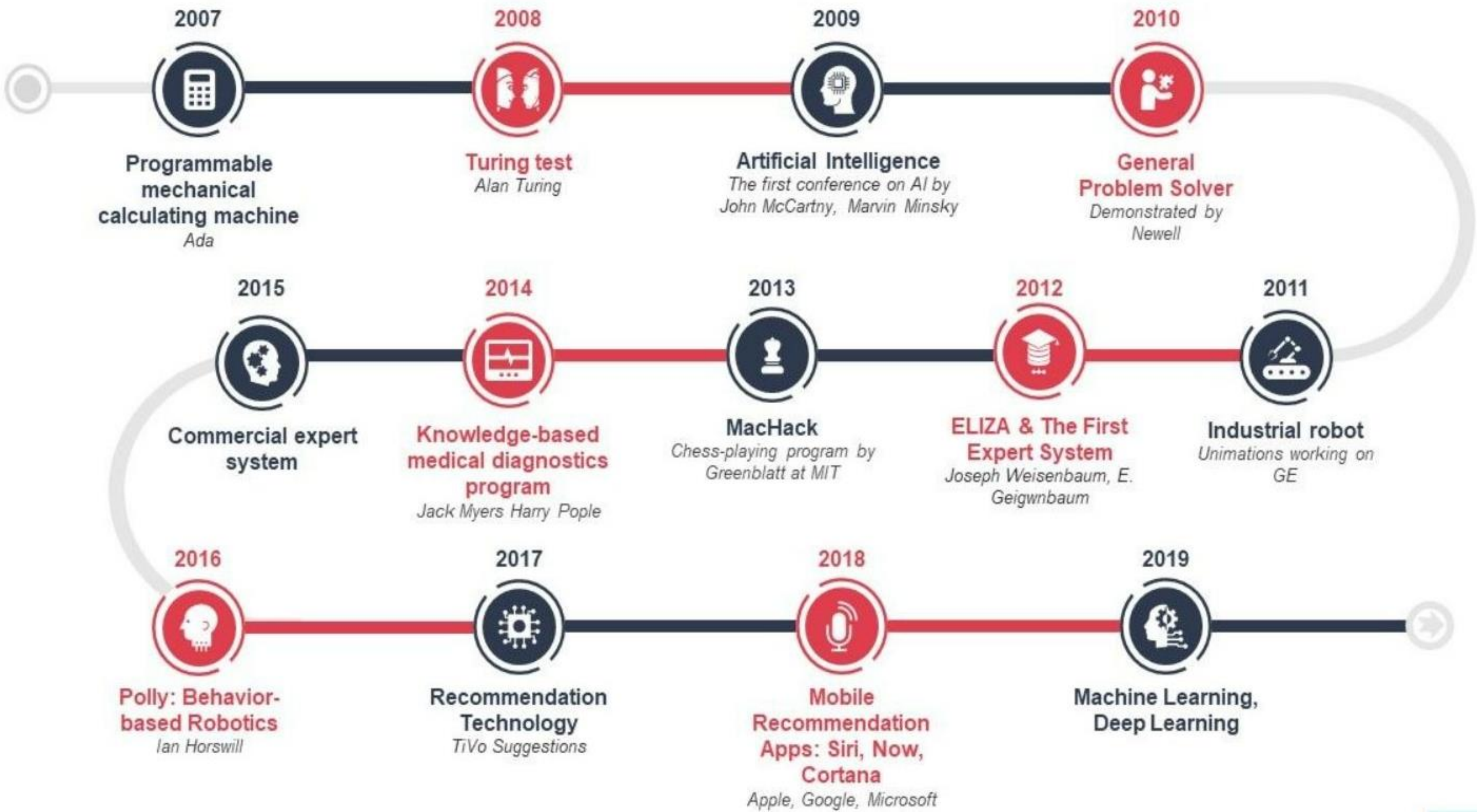
ARTIFICIAL INTELLIGENCE IN DEEP LEARNING / GENERATIVE / AGENTIC AI MODELS

What is AI?

Artificial Intelligence

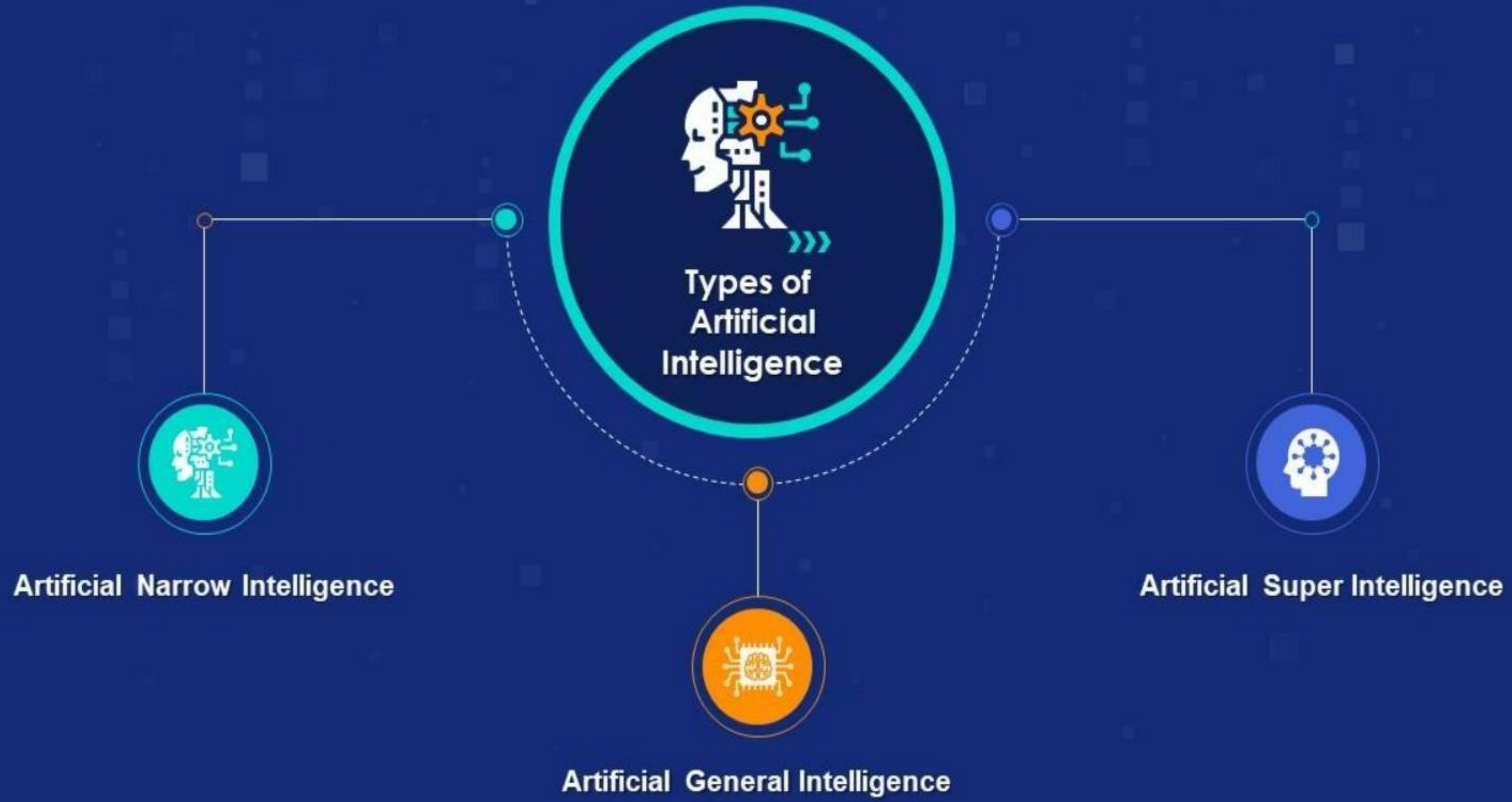
- (AI) is a popular branch of computer science that concerns with building "intelligent" smart machines capable of performing intelligent tasks.
- With rapid advancements in deep learning and machine learning, the tech industry is transforming radically.

Evolution of Artificial Intelligence (AI)



Where is AI used?

Introduction to AI Levels?



AI VS Machine Learning VS Deep Learning

Artificial Intelligence

- Artificial Intelligence originated around 1950s
- AI represents simulate intelligence in machines
- AI is a subset of data science
- Aim is to build machines which are capable of thinking like humans

Machine Learning

- Machine Learning originated around 1960s
- Machine learning is the practice of getting machines to make decisions without being programmed
- Machine learning is a subset of AI & Data Science
- Aim is to make machines learn through data so that they can solve problems

Deep Learning

- Deep Learning originated around 1970s
- Deep Learning is the process of using artificial neural networks to solve complex problems
- Deep Learning is a subset of Machine Learning, AI & Data Science
- Aim is to build neural networks that au6tonetically discover patterns for feature detection

GENERATIVE AI

- **Generative AI** is a type of Artificial Intelligence to create new content, like text, images, Audios and Videos based on patterns in data.
- **Generative AI** typically uses machine learning models, especially deep learning models, to learn from input data and then generate new data based on the patterns and trends it has learned

Key Application:

- Text Generation [ChatGPT, Gemini, Deep Seek]
- Image Generation [DALL-E, Midjourney]
- Video & Speech Generation [Deep Brain, Synthesia]
- Data Augmentation [Synthesis AI]

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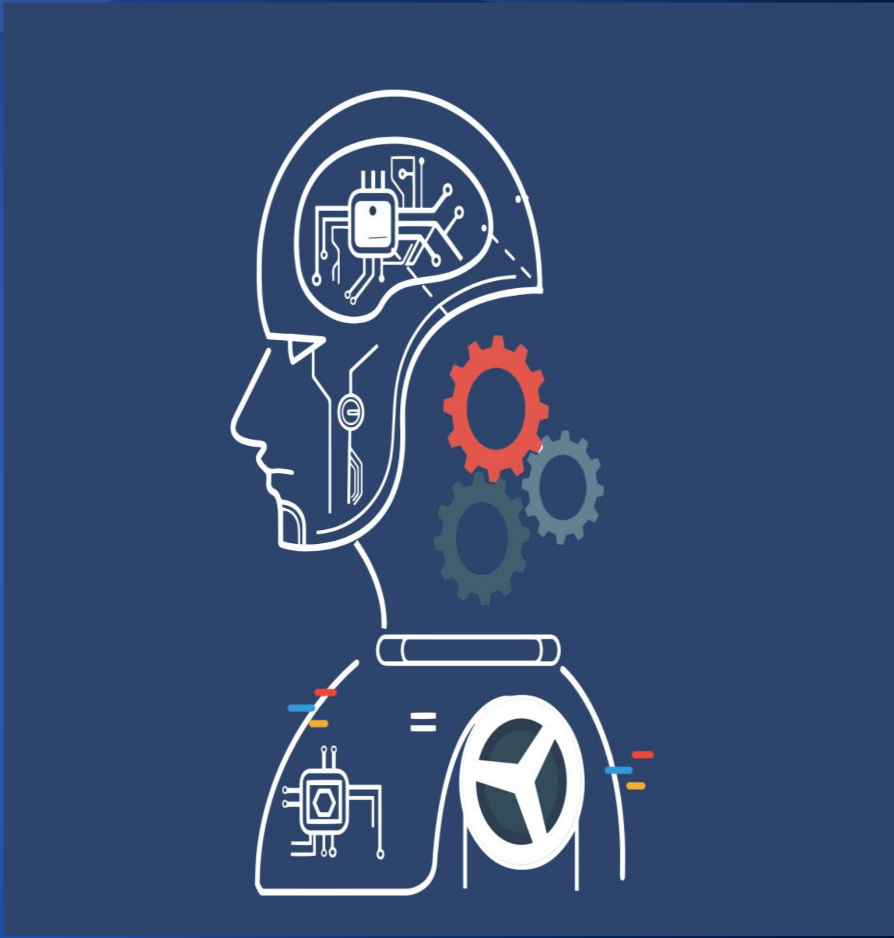
GENERATIVE AI

AGENTIC AI

- **Agentic AI** refers to AI systems designed to act autonomously, making decisions and taking actions with minimal human supervision to achieve specific goals.
- **Agentic AI** is emerging as a transformative technology, particularly with the integration of Large Language Models (LLMs).
- **Agentic AI** systems utilize AI agents, which are specialized software components that can operate independently or cooperatively to solve problems.

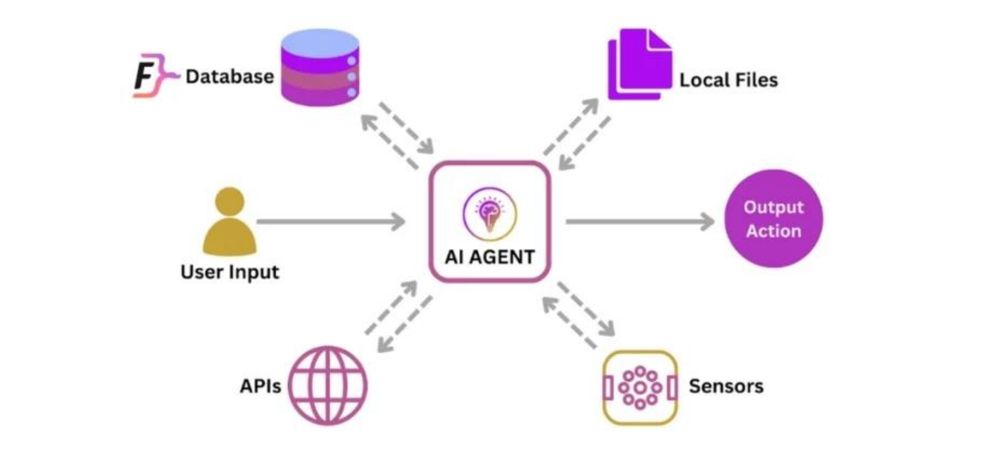


AI AGENT

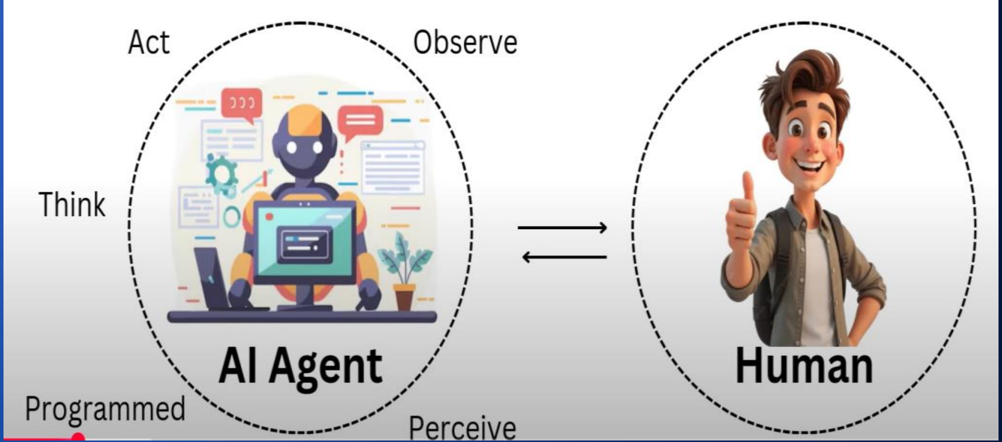
AI Agents are software programs that use artificial intelligence to perform tasks and achieve goals autonomously, often without direct human oversight.

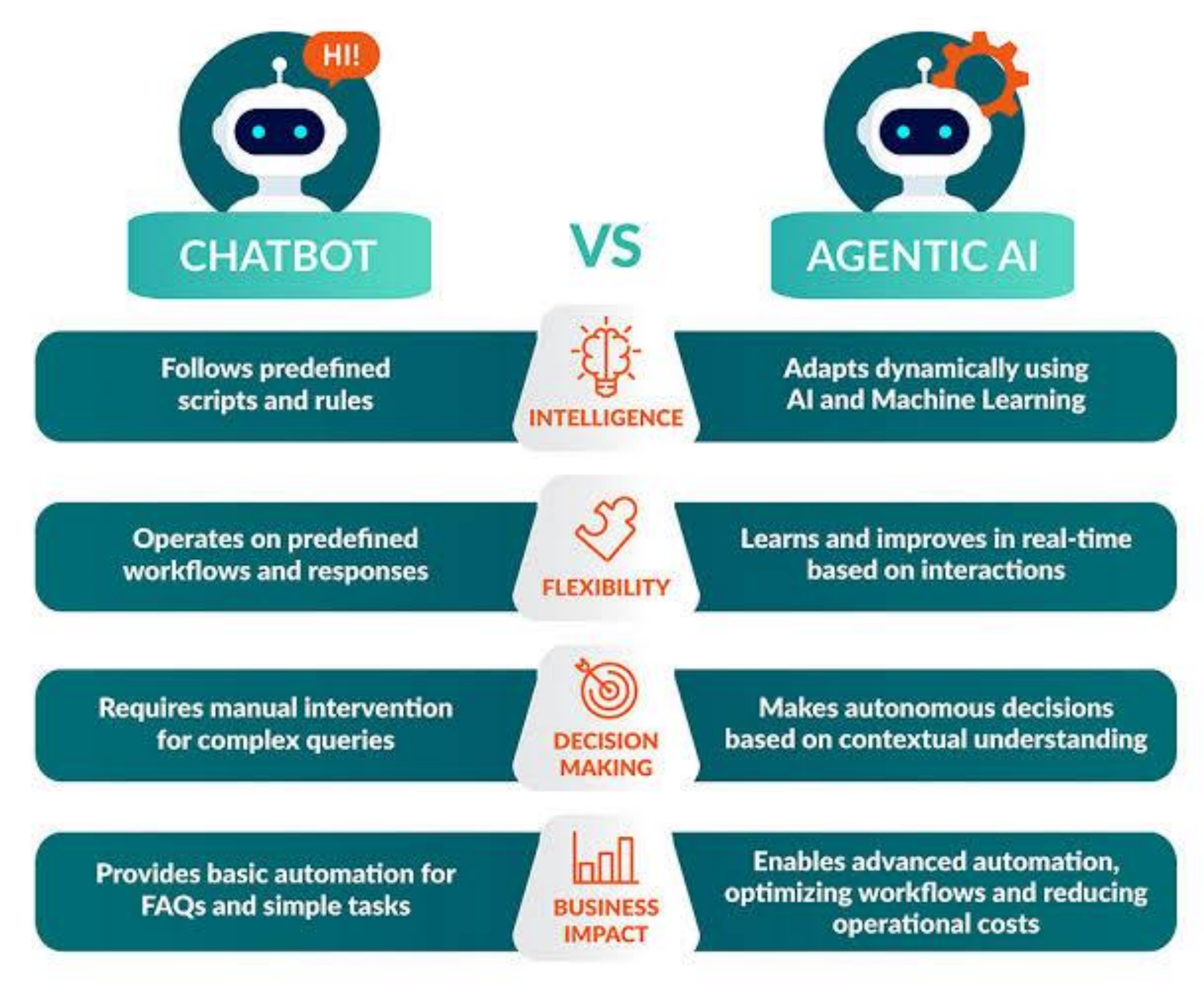
Key Characteristics of AI Agents:

- **Autonomy**
- **Goal-Oriented**
- **Perception**
- **Reasoning & Design Making**
- **Learning & Adaption**
- **Multi-Model Capabilities**
- **Complex Task Handling**



AI Agent Architecture





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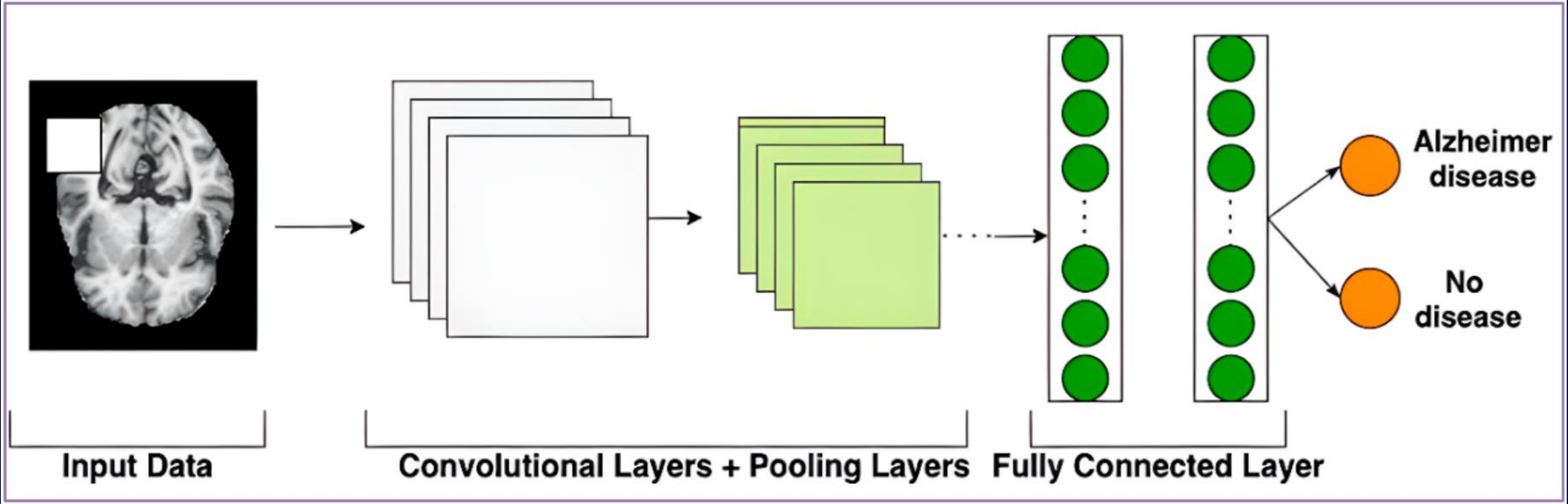
DEEP LEARNING MODELS



CNN MODEL ARCHITECTURE

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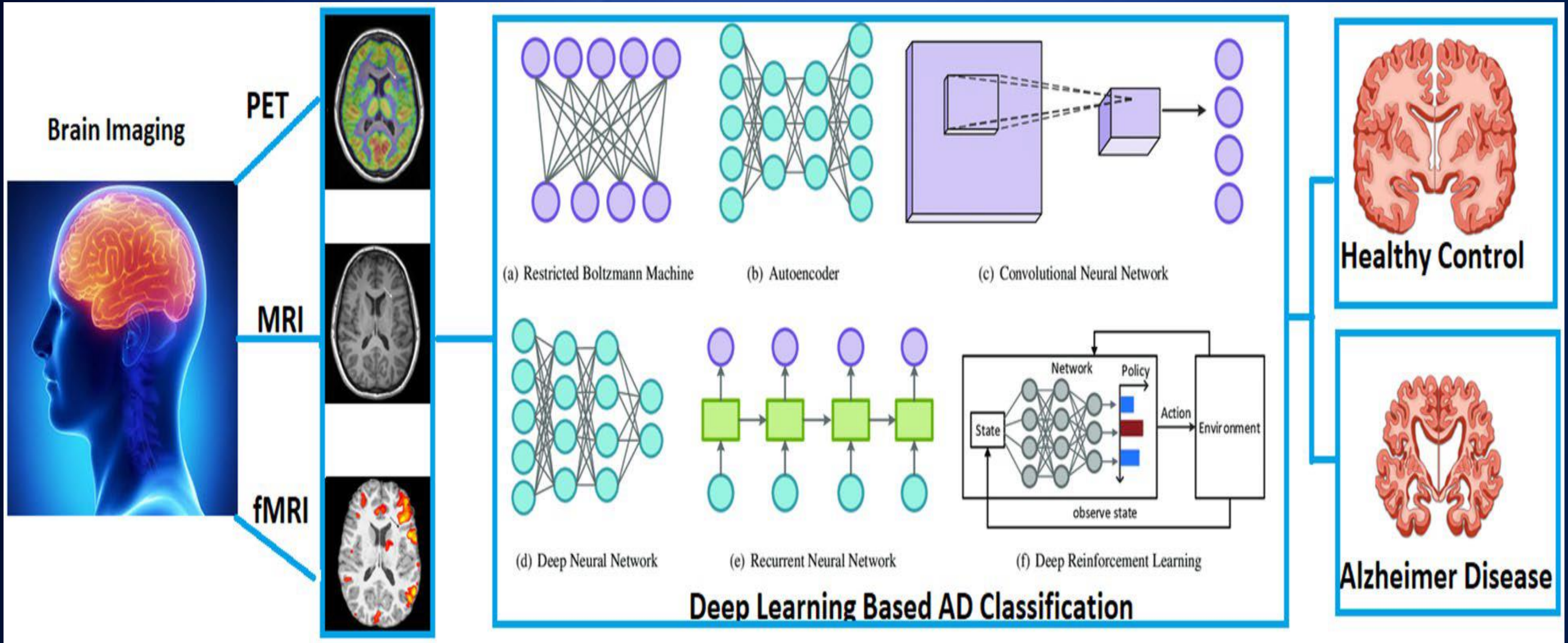
DEEP LEARNING ARCHITECTURE FOR NEURO DISEASES MODELS



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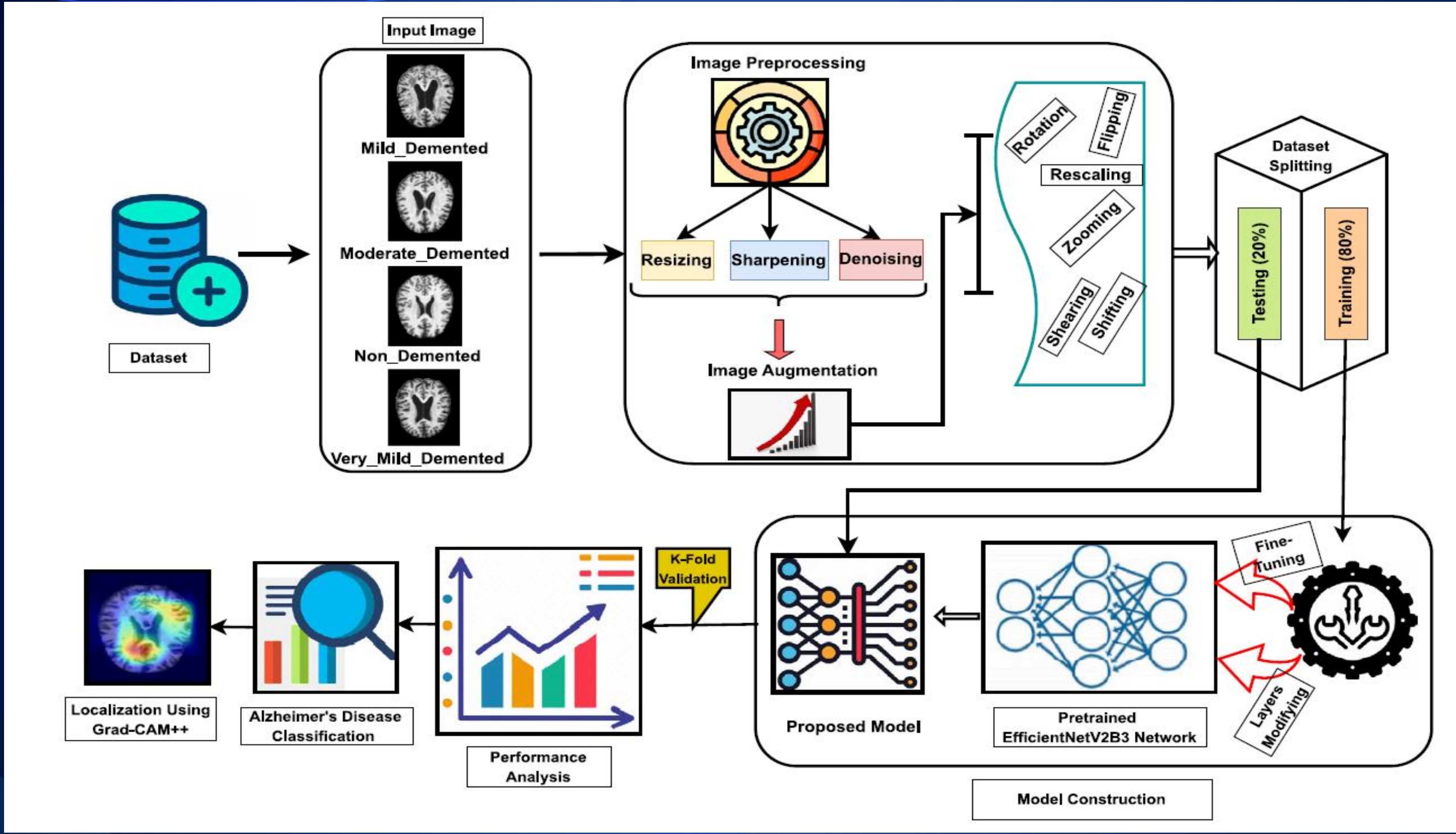
DEEP LEARNING ARCHITECTURE FOR NEURO DISEASES MODELS



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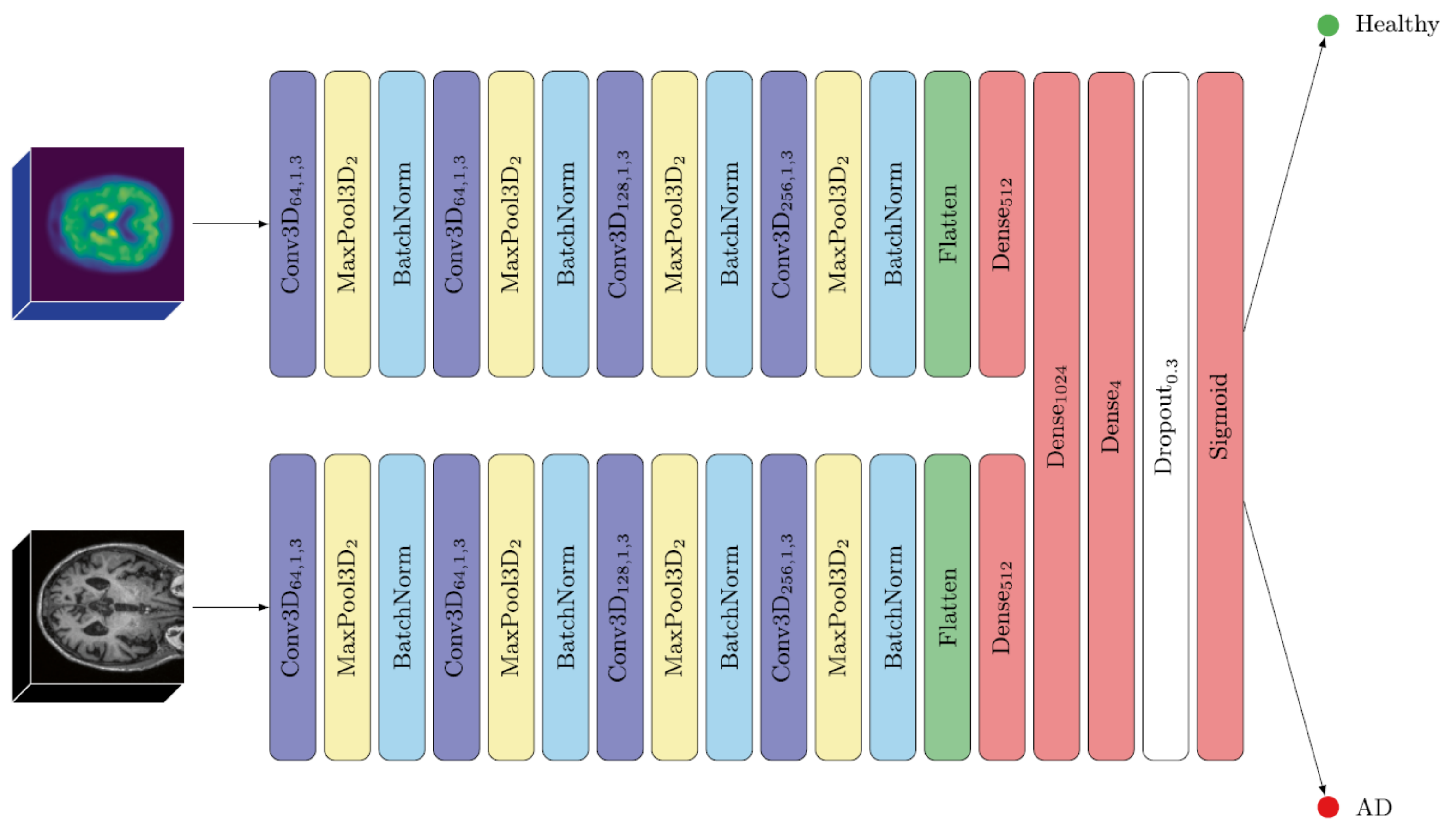
DEEP LEARNING ARCHITECTURE FOR NEURO DISEASES MODELS



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RESEARCH PROSPECTIVES IN NEUROIMAGING [MODELS / XAI / AGENTS]

- **Build Machine / Deep Learning Models / Frameworks for the classification / interpretation of Neuroimaging Disease**

- **Comparison of Multi Cohorts Datasets for the classification / interpretation of Neuroimaging Disease**

- **Build Single / Multi Models For taking Single / Multi modalities**

- **Simple / Systematic Review of Particular Neuroimaging Models / Frameworks / Disease.**

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Comprehensive Overview of Deep Radiomics in Neuroimaging For AD Diagnosis

Modality	Biomarker/Feature	Purpose/Application in AD	Deep Radiomics Benefit
Structural MRI (sMRI)	Brain atrophy	Measures hippocampal and medial temporal lobe atrophy	Detects minute volumetric changes using voxel-based analysis.
Functional MRI (fMRI)	Neuronal connectivity	Assesses activity and connectivity changes in AD	Identifies microstructural abnormalities via texture analysis
Diffusion Tensor Imaging (DTI)	White matter integrity	Evaluates microstructural changes in neuronal pathways	Combines MRI with PET for enhanced diagnostic accuracy
Amyloid PET	Amyloid-beta	Identifies amyloid plaque accumulation	Enhances early diagnosis by detecting AD before clinical symptoms appear
Tau PET	Tau protein	Correlates with neurodegeneration and cognitive decline	Assists in predicting MCI to AD conversion
FDG-PET	Glucose metabolism	Detects hypometabolism in AD-prone regions	Identifies subtle imaging patterns via automated feature extraction

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RESEARCH QUESTIONS & MOTIVATION FOR ALZHEIMER'S DISEASE DIAGNOSIS

SrNo	Research Questions	Motivation
RQ1	What multimodal data combinations are used in the detection of AD?	To know how complimentary modalities improve prediction and characterization
RQ2	What datasets and preprocessing methods are used for multimodal data in Alzheimer's Disease research?	To understand the what preprocessing methods are essential for ensuring data compatibility, reproducibility, and model performance
RQ3	What methods are used for multimodal data fusion in Alzheimer's Disease detection?	By reviewing the fusion methods enables assessment and their effectiveness and suitability for AD diagnosis
RQ4	What different AI models are available for AD diagnosis that incorporates multimodal data?	To identify the best practices and approaches for effectively learning from complex multimodal representation
RQ5	Which XAI frameworks are used to interpret AI models for Alzheimer's Disease using multimodal data?	To explore how XAI techniques enhance trust, transparency, and model validation in a multimodal setting
RQ6	What are the performance trends of AI models in multimodal Alzheimer's Disease diagnosis across different datasets, fusion strategies, and classifiers?	To know the extent to which multimodal fusion and interpretability improve diagnostic performance over traditional methods
RQ7	What are the limitations, challenges, and future scope of using multimodal data and XAI methods for AD diagnosis?	To provide a direction for future research and help to overcome real-world barriers in adoption and generalization

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